CUBAN NATIONAL CHEMISTRY CONTEST

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"To work in the sterile when one can do the useful; to deal with the easy when one has spirit to try the difficult, is to strip the dignity from the talent."

José Martí

Name(s): ______________________________ Last name(s): __________________________

School: ____

Municipality: ___________________________ Province: ______________________________

CODE: ________________________________ Test: **DAY 2. TWELFTH GRADE EXAM**

Instructions

- Write clearly all your data in the appropriate place (names, last names, school, municipality, province). Do not write anything in the field CODE.
- Verify that your questions booklet is complete. This is the Day 2 problem set, with an individual exam for 12^{th} grade. The exam has 3 problems (problem 11, 10 items; problem 12, 4 items; problem 13, 11 items) and a total of 6 pages.
- All data, constants, conversions and basic equations needed are included in the exam. You are allowed to use a scientific calculator.
- At the beginning of each problem its scoring (from a total of 100 points), each item scoring (in marks) is indicated.

Success!

Ministerio de Educación

República de Cuba

FACULTAD DE QUÍMICA

UNIVERSIDAD DE LA HABANA

CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's Constant	$N_A = 6{,}0221 \cdot 10^{23}\ mol^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gasses' constant	$R = 8{,}3145\ J \cdot K^{-1} \cdot mol^{-1}$	Thermodynamics First Principle	$Q = \Delta E + W$
Cero in the Celsius scale	$273{,}15\ K$	Gibbs' energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485\ C \cdot mol^{-1}$	Nernst's equation	$E = E° - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{red}}{c_{ox}}$
Water ionic product at 25 °C	$K_W = 1{,}00 \cdot 10^{-14}$	Arrhenius' equation	$k = A \cdot \exp\left(-\frac{E_A}{R \cdot T}\right)$
Standard conditions	$p° = 100\ kPa$ $c° = 1\ mol \cdot L^{-1}$	$\Delta G° = -R \cdot T \cdot \ln K° = -z \cdot F \cdot \Delta E°$	
$1\ J = 1\ C \cdot 1\ V = 1\ kPa \cdot 1\ L = 1\ W \cdot 1\ s$		$1\ C = 1\ A \cdot 1\ s$	

PERIODIC TABLE OF THE ELEMENTS

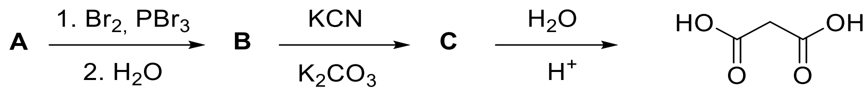
1 1A	2 2A	3 3B	4 4B	5 5B	6 6B	7 7B	8 8B	9 8B	10 8B	11 1B	12 2B	13 3A	14 4A	15 5A	16 6A	17 7A	18 8A
1 H 1.008																	2 He 4.003
3 Li 6.941	4 Be 9.012											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	57 La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (263)	107 Bh (262)	108 Hs (265)	109 Mt (266)	110 Ds (269)	111 Rg (272)	112 Uub (277)		114 Uuq (2??)		116 Uuh (2??)		118 Uuo (2??)

58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Problem 11. Organic Synthesis (20 points)

11.1	11.2	11.3	11.4	11.5	11.6	11.7	11.8	11.9	11.10	Total	Total
6	1	7	4	2	8	4	3	2	3	40	20

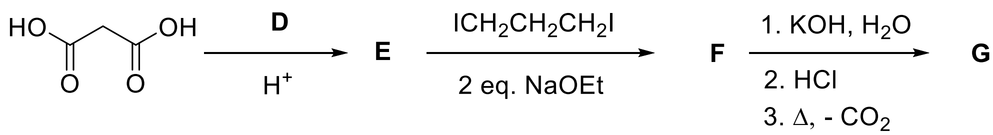
Malonic acid can be obtained according to the following.



11.1 Draw the structures of compounds **A**, **B** and **C**.

11.2 Which is the name of the first reaction in the sequence?

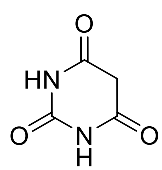
Malonic synthesis allows the obtaining of monocarboxylic acids of great molecular complexity. An example is shown in the following.



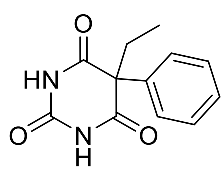
11.3. Draw the structures of compounds **D**, **E**, **F** and **G**.

11.4. Propose a mechanism for all the reactions that occur in the last step of the previous sequence.

Malonic acid derivatives have been employed in the obtention of barbituric acids, drugs that act as sedatives of the central nervous system and produced various effects. Phenobarbital, manufactured by Bayer company under the brand Luminal®, is sued as an anticonvulsant.



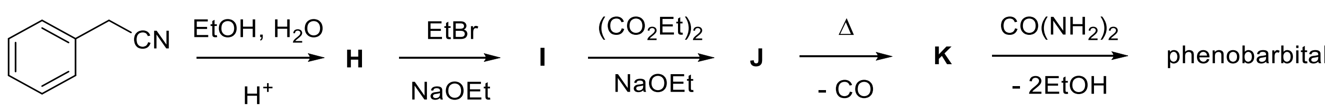
barbituric acid



phenobarbital

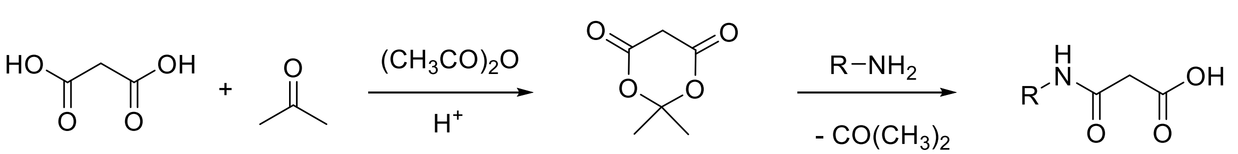
11.5. Indicate the most acidic hydrogens of barbituric acid.

Phenobarbital can be obtained from benzyl cyanide according to the following sequence.



11.6. Draw the structures of compounds **H**, **I**, **J** and **K**.

Meldrum's acid can be obtained by the reaction between malonic acid and propanone. Its reaction with amines allows for getting synthetic intermediates of high utility.



Meldrum' acid

11.7. Propose a mechanism for the formation of Meldrum's acid.

11.8. Propose a mechanism for the reaction between Meldrum's acid and a primary amine.

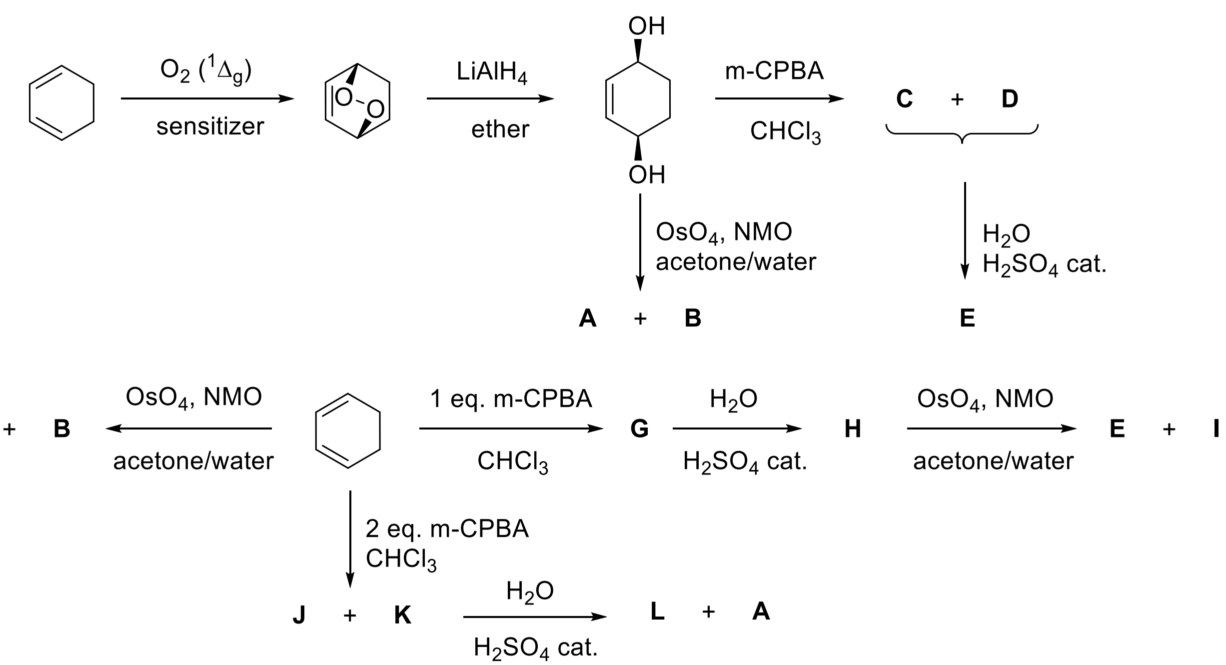
11.9. Which thermodynamic factors favor the previous reaction?

11.10. Draw the resonance structures of the enolate derived from Meldrum's acid.

Problem 12. Stereochemistry (15 points)

12.1	12.2	12.3	12.4	Total	Total
24	1	3	2	30	15

Recently cyclitols have attracted attention due to their diverse biological activities and versatility as synthetic intermediates. Polyhydroxycycohexanes such as inositols, quercitols and conduritols belong to the family of cyclitols. These compounds can have several stereoisomers. cyclohexa-1,3-diene is an important compound for the synthesis of cyclitols derivatives. The scheme below shows the total synthesis of several isomeric cyclitols with formula $C_6H_{12}O_4$ (compounds **A**, **B**, **E**, **F**, **I** and **L**). In the case of chiral compounds only enantiomers are represented.



m-CPBA: meta-chloroperbenzoic acid

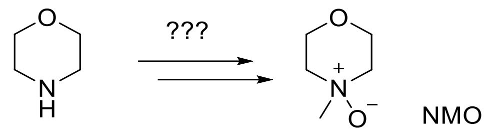
NMO: N-methylmorpholine N-oxide. Oxidant used to regenerate OsO_4 consumed during the reaction.

12.1. Draw the structures of compounds **A-L** knowing that compounds **E**, **F**, **G**, **H**, **I**, **J** and **L** are optically actives.

12.2. Tell the name of the reaction between cyclohexa-1,3-diene and singlet dioxygen.

12.3. Represent the chair conformations of compound **I**. Tell which is the most stable.

12.4. How can you obtain NMO starting from morpholine?



Problem 13. Precipitation equilibria (15 points)

13.1	13.2	13.3	13.4	13.5	13.6	13.7	13.8	13.9	13.10	13.11	Total	Total
2	2	2	4	1	1	4	1	4	5	4	30	15

The solubility product constant describes the equilibrium established when a sparingly soluble electrolyte is dissolved in water. There are several methods to determine the value of this constant. In the following discussion three examples are presented.

A- Calcium oxide was added to pure water and the resultant suspension was allowed to rest for a week in a plastic and closed container under an inert atmosphere. The suspension was filtered to eliminate the solid generated and transferred for analysis. Immediately, several 20,00 mL aliquots of the filtered solution were titrated with hydrochloric acid, previously standardized and with a concentration of 0,0150 $mol \cdot L^{-1}$, in the presence of phenolphthalein as an acid-base indicator. The volumes consumed from the titrant were 30,5, 29,4, 29,8 y 29,6 mL.

13.1. In the previously described procedure, several precautions were followed to avoid introducing errors. Why is it necessary to store the problem solution in a plastic container and under an inert atmosphere? Why was the mixture allowed to stand for a week?

13.2. Describe the principle behind the functioning of an acid-base indicator.

13.3. Before titration, it is necessary to standardize the hydrochloric acid because this substance is not a primary standard. Which characteristics possess these substances?

13.4. Calculate the solubility product constant of calcium hydroxide.

B- A 100,0 mL aliquot of a saturated solution of lead(II) chloride was reacted with 50,0 mL of 2,00 $mol \cdot L^{-1}$ potassium sulfate solution. The mixture was allowed to stand under an inert atmosphere. Afterward, the mixture was filtered and the precipitate was washed with cold water, dried under heating and weighed until reaching constant mass. The previous procedure was repeated, obtaining 0,4821, 0,4822, 0,4827 and 0,4825 g of the precipitate.

13.5. Why is it necessary to heat the precipitate?

13.6. Which is the name of the method employed in this case?

13.7. Calculate the solubility product constant of lead(II) chloride.
$M(PbSO_4) = 303{,}26\ mol \cdot L^{-1}$, $K_{ps}\ (PbSO_4) = 1{,}6 \cdot 10^{-8}$.

C- Different volumes of solutions of silver nitrate 0,100 $mol \cdot L^{-1}$ and potassium chloride 0,100 $mol \cdot L^{-1}$ were mixed in a beaker. The resultant solutions were stirred for 15 minutes and the electrode potential was measured. For the determination, a digital voltmeter was employed, a silver electrode (in the cathode) was the indicator electrode and a saturated Calomel electrode ($E^o(Hg_2Cl_2/Hg) = 0{,}241$ V) was used as the reference electrode. To conduct the measurement, the indicator electrode was immersed in the solution and the reference electrode was immersed in a

Luggin capillary full of a saturated solution of potassium nitrate (salt bridge) in contact with the solution. The table shows the results of the cell potential obtained at 25 °C.

Mezcla	$V(AgNO_3)$/mL	V(KCl)/mL	ΔE/V
1	100	0	0,499
2	90	10	0,493
3	80	20	0,486
4	70	30	0,475
5	60	40	0,457
6	50	50	0,269
7	40	60	0,082
8	30	70	0,064
9	20	80	0,057
10	10	90	0,046

13.8. Why cannot the reference electrode directly contact the solution?

13.9. Find the linear equations that express the dependency of the cell potential when $AgNO_3$ is in excess (ΔE *vs* ln $c(Ag^+)$) and when KCl is in excess (ΔE *vs* ln $c(Cl^-)$).

13.10. Calculate the standard electrode potentials E°(AgCl/Ag) and E°(Ag^+/Ag).

If you cannot solve the previous item, use the values E°(AgCl/Ag) = 0,72 V and E°(Ag^+/Ag) = 1,30 V for the standard electrode potential when solving the rest of the problem.

13.11. Calculate the solubility product constant of silver chloride.